

A double integral that is constant on symmetry orbits

two involutions of the disk flatten the integrand to $\pi/8$

$$\boxed{\iint_{x^2+y^2 \leq 1} \frac{\arctan \frac{1+x^2}{1+y^2}}{1+e^{-y}} dx dy = \frac{\pi^2}{8}.}$$

Overview. The integrand looks unpleasant: a logistic weight in y alone, multiplying an arctangent of a ratio. It is never evaluated. Instead one observes that the disk carries four measure-preserving maps under which the integral is invariant, and that the sum of the four transported copies of the integrand is the constant $\pi/2$. Averaging, the integrand may be replaced by $\pi/8$ without changing the value, and the answer is $\pi/8$ times the area of the disk.

Two elementary identities carry the whole argument:

$$\sigma(t) + \sigma(-t) = 1, \quad \sigma(t) = \frac{1}{1+e^{-t}}, \quad (1)$$

$$\arctan u + \arctan \frac{1}{u} = \frac{\pi}{2}, \quad u > 0. \quad (2)$$

The first kills the logistic factor, the second kills the arctangent.

1. Notation and the symmetry group

Write $D = \{(x, y) : x^2 + y^2 \leq 1\}$ and

$$A(x, y) = \arctan \frac{1+x^2}{1+y^2}, \quad \sigma(t) = \frac{1}{1+e^{-t}}, \quad g(x, y) = A(x, y) \sigma(y),$$

so that $I = \iint_D g$.

Consider the four maps of the plane

$$T_1(x, y) = (x, y), \quad T_2(x, y) = (x, -y), \quad T_3(x, y) = (y, x), \quad T_4(x, y) = (y, -x). \quad (3)$$

Each is linear with determinant ± 1 , hence preserves Lebesgue measure, and each preserves $x^2 + y^2$, hence maps D bijectively onto itself. T_2 is the reflection in the x -axis, T_3 the reflection in the diagonal $y = x$, and $T_4 = T_2 T_3$ a quarter turn.

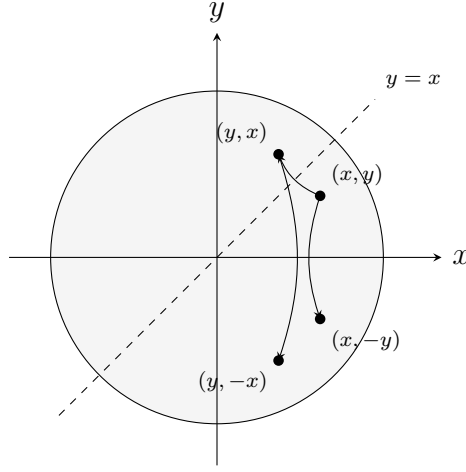
Consequently, for every j ,

$$\iint_D g(T_j(x, y)) dx dy = \iint_D g(x, y) dx dy = I,$$

and averaging the four identities,

$$I = \frac{1}{4} \iint_D [g(x, y) + g(x, -y) + g(y, x) + g(y, -x)] dx dy. \quad (4)$$

No property of g has been used so far; (4) holds for any integrable g .



the four points share one value of the average, namely $\pi/8$

2. The logistic factor pairs off

The function A depends on y only through y^2 , so $A(x, -y) = A(x, y)$ and $A(y, -x) = A(y, x)$. Using (1) twice,

$$g(x, y) + g(x, -y) = A(x, y)[\sigma(y) + \sigma(-y)] = A(x, y), \quad (5)$$

$$g(y, x) + g(y, -x) = A(y, x)[\sigma(x) + \sigma(-x)] = A(y, x). \quad (6)$$

The logistic weight has vanished from the problem. Its only relevant property was

$$\sigma(t) + \sigma(-t) = \frac{1}{1 + e^{-t}} + \frac{1}{1 + e^t} = \frac{1 + e^t + 1 + e^{-t}}{(1 + e^{-t})(1 + e^t)} = 1,$$

since $(1 + e^{-t})(1 + e^t) = 2 + e^t + e^{-t}$.

3. The arctangent pairs off

Put

$$u = \frac{1 + x^2}{1 + y^2} > 0, \quad \text{so that} \quad \frac{1 + y^2}{1 + x^2} = \frac{1}{u}.$$

Then $A(x, y) = \arctan u$ and $A(y, x) = \arctan \frac{1}{u}$, and (2) gives

$$A(x, y) + A(y, x) = \frac{\pi}{2}. \quad (7)$$

Identity (2) itself is immediate: for $u > 0$ the angles $\alpha = \arctan u$ and $\beta = \arctan \frac{1}{u}$ are the two acute angles of the right triangle with legs 1 and u , so $\alpha + \beta = \pi/2$. Analytically, the derivative of $\arctan u + \arctan \frac{1}{u}$ is $\frac{1}{1+u^2} - \frac{1/u^2}{1+1/u^2} = 0$ on $(0, \infty)$, and the value at $u = 1$ is $\frac{\pi}{4} + \frac{\pi}{4}$.

Adding (5) and (6) and inserting (7),

$$\boxed{g(x, y) + g(x, -y) + g(y, x) + g(y, -x) = \frac{\pi}{2} \quad \text{for all } (x, y).} \quad (8)$$

This is a pointwise identity, valid on the whole plane, not merely in the mean.

4. Conclusion

Substituting (8) into (4), the bracket is the constant $\pi/2$, so

$$I = \frac{1}{4} \iint_D \frac{\pi}{2} dx dy = \frac{\pi}{8} \text{area}(D) = \frac{\pi}{8} \cdot \pi = \frac{\pi^2}{8}.$$

The geometric reading is that g itself is a genuinely lopsided surface over the disk, larger where $y > 0$ because of the logistic weight, but the four-point symmetry orbit of any point carries a fixed total height $\pi/2$. Integration only sees orbit averages, and every orbit average equals $\pi/8$.

5. What the problem really requires

Each ingredient can be replaced. Here is exactly what is used.

Ingredient	Property actually used
The domain D	Invariance under $(x, y) \mapsto (x, -y)$ and $(x, y) \mapsto (y, x)$. Any such region works: a disk, a square $[-a, a]^2$, an annulus, the whole plane with an integrable weight.
The weight $\sigma(y) = (1 + e^{-y})^{-1}$	Only $\sigma(t) + \sigma(-t) = 1$. Equivalently $\sigma = \frac{1}{2} + \text{odd}$. Examples: $\frac{1}{2}(1 + \tanh y)$, $\frac{1}{2} + \frac{\arctan y}{\pi}$, $\frac{1}{2} + y^3 e^{-y^2}$.
The ratio $\frac{1+x^2}{1+y^2}$	Only that it is positive, even in each variable separately, and inverted by the swap $x \leftrightarrow y$. Examples: $\frac{a+x^4}{a+y^4}$, $e^{x^2-y^2}$, $\frac{\cosh x}{\cosh y}$.
$\arctan$	Only the reflection law $\varphi(u) + \varphi(1/u) = \text{const.}$

This yields the general statement.

Proposition. Let $D \subset \mathbb{R}^2$ have finite area and be invariant under $(x, y) \mapsto (x, -y)$ and $(x, y) \mapsto (y, x)$. Let $h > 0$ be even in each variable separately with $h(x, y)h(y, x) = 1$, and let σ satisfy $\sigma(t) + \sigma(-t) = 1$. Then

$$\boxed{\iint_D \arctan(h(x, y)) \sigma(y) dx dy = \frac{\pi}{8} \text{area}(D).} \quad (9)$$

Proof. Identical to Sections 2–4: average over $T_1, \dots, T_4$, use $\sigma(t) + \sigma(-t) = 1$ to remove σ , then $\arctan h + \arctan h^{-1} = \pi/2$. $\square$

6. Why the direct approach fails

The natural first move, integrating in y for fixed x or passing to polar coordinates, fails: neither the logistic factor nor the arctangent has an elementary antiderivative in this combination, and polar coordinates destroy the $x \leftrightarrow y$ structure by mixing both variables into θ . The question to ask is which maps preserve the domain, and what the integrand does under them. Here the integrand is not invariant under any single one of those maps, which is why the symmetrisation (4), an average rather than an identification, is the right instrument.

Numerically, adaptive quadrature over the disk gives $1.2337005\dots$, against $\pi^2/8 = 1.2337005501\dots$, consistent to the accuracy of the quadrature.